

- 29** In a therapy unit, patients are given treatment from a certain radioactive source. This source has a half-life of 4 years. A particular treatment requires 10 minutes of irradiation when the source is first used.

How much time is required for this treatment, using the same source, 2 years later?

- A** 7 minutes **B** 12 minutes **C** 14 minutes **D** 20 minutes

Ans: C

$$A_{2\text{years}} = \frac{1}{2}^{0.5} A_0$$

$$A \propto \frac{1}{t}$$

and thus $t_{2\text{years}} = 14\text{mins}$

20. The rest masses of three particles, tritium ${}^3\text{H}$, proton ${}^1\text{p}$, and neutron ${}^1\text{n}$, are 3.0160 u